B4.2 Ecological niches

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-b4-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 363–379 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

NICHES AND OXYGEN

B4.2.1 A niche is a species' role in an ecosystem

pp. 363–364

A **niche** includes the biotic and abiotic interactions that influence growth, survival and reproduction, including how a species obtains food.

B4.2.2 Obligate and facultative groups

pp. 364–365

GROUP	OXYGEN	EXAMPLE
OBLIGATE AEROBE	Needed	Most animals and plants
FACULTATIVE ANAEROBE	Tolerated, but not needed	Yeast, *E. coli*
OBLIGATE ANAEROBE	Not tolerated	*Clostridium tetani*

MODES OF NUTRITION

B4.2.3 Photosynthesis

p. 366

Plants, eukaryotic algae and several groups of prokaryotes, including cyanobacteria, photosynthesize.

B4.2.4 Holozoic nutrition

pp. 366–367

All animals are heterotrophs, and most are holozoic. Food is **ingested**, **digested** internally, **absorbed** and **assimilated**.

B4.2.5 Mixotrophic nutrition

p. 367

Mixotrophs both photosynthesize and feed. *Euglena* is a freshwater example, and many others live in ocean plankton. Facultative mixotrophs can use either mode, whereas obligate ones need both.

B4.2.6 Saprotrophic nutrition

pp. 367–368

Many fungi and bacteria secrete enzymes onto dead matter and absorb the products. They are also called **decomposers**.

B4.2.7 Diversity of nutrition in archaea

pp. 368–369

Archaea are one of the three domains of life. They include photosynthetic, chemosynthetic and heterotrophic organisms, and many live in extreme environments.

FEEDING AND BEING EATEN

B4.2.8 Teeth and diet in the Hominidae

pp. 369–372

Teeth show diet. *Paranthropus robustus* had very large, flat molars and strong jaws, suited to tough plants. *Homo sapiens* has small, flat teeth that suit a mixed diet. *Homo floresiensis* had large premolars and a strong jaw, and tooth wear suggests a mainly tough plant diet with some meat.

B4.2.9 Herbivores and plant defences

pp. 372–373

Leaf-eating insects have chewing or piercing mouthparts. Plants resist with spines, stings and toxic secondary compounds in seeds and leaves. Some herbivores have metabolic adaptations that detoxify these substances: the proboscis monkey has gut bacteria that neutralize toxins in the leaves it eats.

B4.2.10 Predators and prey

pp. 373–376

Predators and prey

B4.2.10



Wolf. NPS, public domain.



Leaf-tailed gecko. C. J. Sharp, CC BY-SA 4.0.

Type	Predators	Prey
Physical	Teeth, claws, keen senses	Camouflage, shells, spines
Chemical	Venom	Toxins, warning colours
Behavioural	Group hunting, ambush	Groups, alarm calls, hiding

Activity

For a local predator and prey, suggest one adaptation of each type.

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Predators and prey both have chemical, physical and behavioural adaptations for finding, catching and killing prey, or for resisting predation.

B4.2.11 How forest plants reach light

pp. 376–377

Trees grow up to the canopy. **Lianas** climb on them, **epiphytes** grow on their branches, and **strangler figs** begin high up and send roots down. Shade-tolerant plants survive in low light on the forest floor.

COMPETITION

B4.2.12 Fundamental and realized niches

pp. 377–378

A **fundamental niche** is the potential of a species, based on adaptations and tolerance limits. A **realized niche** is its actual extent when it competes with other species.

B4.2.13 Competitive exclusion

pp. 378–379

Each species has a unique niche. If two species compete, one may be eliminated, or both may be restricted to part of their fundamental niche. Gause showed this in 1934: grown together, *Paramecium aurelia* excluded *P. caudatum*.

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